



CYBER SECURITY

Workshop Guide

Stay Safe Online!

TOPICS COVERED

1. Data Privacy & GDPR
2. Phishing
3. Viruses & Malware
4. Cryptography & HTTPS
5. VPN (Virtual Private Network)
6. Passwords, MFA & Passkeys
7. AI & Deepfakes
8. App vs Browser

For ages 10 – 16 | Cybersecurity Workshop



DATA PRIVACY

Your personal data is precious — protect it!

What is Data Privacy?

Every time you use an app or website, you share personal information: your name, age, photos, location, and more. Data privacy means having control over WHO can see and USE that data.

EU GDPR — The European Data Protection Law

The GDPR (General Data Protection Regulation) is a European law that protects your personal data. It gives you rights:

- Right to know: companies must tell you what data they collect
- Right to access: you can ask to see your own data
- Right to delete: you can ask companies to erase your data
- Right to say NO: you can refuse certain uses of your data

Even if you live outside Europe, many websites apply GDPR rules globally!

Good Habits

Read privacy settings in apps
Use a nickname in games
Ask a parent before signing up
Avoid sharing your address online

Watch Out!

Sharing your full name + school online
Accepting all cookies without reading
Posting your exact location publicly
Using real birthday in all apps



Did you know?

Every 'free' app or service often pays for itself by collecting and selling your data to advertisers. If it's free... YOU might be the product!



Think About It!

What personal information do you share online without thinking? Name 3 things you could stop sharing today.



PHISHING

When scammers pretend to be someone you trust

What is Phishing?

Phishing is when a scammer sends you a fake email, message, or link pretending to be a trusted source (your bank, school, or even a friend). The goal is to trick you into giving away your password, credit card number, or personal info.

How to Spot a Phishing Attack



Suspicious sender

The email address looks weird: 'support@paypa1.com' (with the number 1 instead of L)



Urgent language

"Your account will be DELETED in 24 hours! Click now!" — scammers create panic to stop you from thinking clearly.



Suspicious links

Hover over any link before clicking — the real URL often looks different from what's shown.



Spelling mistakes

Real companies have professional communication. Lots of typos = red flag!



If you receive a suspicious message...

- Do NOT click any links
- Do NOT reply with personal info
- Delete the message
- Tell a parent or teacher



What to do

- Report it as spam/phishing
- Contact the real company directly
- Warn your friends if needed
- Trust your instincts!

<https://safeonweb.be/>



Think About It!

You get an email saying: 'You won a free iPhone! Click here to claim it.' What are 3 clues that this might be phishing?



VIRUSES & MALWARE

Sneaky software designed to cause problems

What is a Computer Virus?

A computer virus is malicious software (malware) that secretly enters your device and can steal data, delete files, spy on you, or use your computer to attack others — all without you knowing!

Virus

Copies itself and spreads to other files, like a real virus spreads between people.

Trojan

Disguises itself as a normal app or game. You install it willingly!

Ransomware

Locks all your files and demands money to unlock them.

How to Stay Protected

- Install antivirus software and keep it updated
- Only download apps from official stores (App Store, Play Store)
- Never open email attachments from unknown senders
- Keep your operating system updated (updates fix security holes!)
- Don't plug in random USB drives you find
- Use different passwords on each service



Fun fact

The first computer virus was created in 1983 as an experiment. Today, over 450,000 new malware programs are detected EVERY DAY!



Think About It!

Your friend sends you a link to a 'free Minecraft hack'. What should you think before clicking? What could go wrong?



CRYPTOGRAPHY

Secret codes that protect your data online



What is Cryptography?

Cryptography is the art of scrambling information so only the right person can read it. Imagine writing a message in a secret code that only your best friend can decode!



A Simple Example: Caesar Cipher

Shift every letter 3 positions forward: A becomes D, B becomes E, C becomes F...

Original: HELLO

Encrypted: KHOOR



HTTPS — Encryption for the Web

When you see HTTPS:// and a padlock in your browser, it means your connection is encrypted. This protects your passwords and data from being intercepted!



HTTP (no S)

No encryption. Anyone on the same WiFi can read your data. AVOID for anything important!



HTTPS

Encrypted! Even if someone intercepts your data, they see only scrambled nonsense.



Real-world crypto

Encryption protects: your WhatsApp messages (end-to-end encryption), your bank transactions, your passwords stored in apps, and government secrets!



Think About It!

You're in a cafe using public WiFi and you log into your bank. The site shows HTTP (no S). What's the risk?



VPN

Virtual Private Network — online anonymity with limits!

What is a VPN?

A VPN creates a private, encrypted tunnel between your device and the internet. It hides your IP address (your digital home address) from websites and masks your location.

How a VPN Works

1

Without VPN

Your device → Website directly. The website sees YOUR real IP address and location.

2

With VPN

Your device → Encrypted tunnel → VPN Server → Website. The website sees the VPN server's IP, not yours.



What a VPN CAN do

Hide your IP from websites
Encrypt your internet traffic
Help access geo-blocked content
Protect you on public WiFi



What a VPN CANNOT do

Make you 100% anonymous
Stop you from being hacked
Protect you from viruses
Remove your digital footprint completely



Important: Your VPN Provider CAN See You!

This is a crucial point many people miss:

When you use a VPN, the VPN company can still see ALL your internet traffic. You're just moving your trust from your internet provider to the VPN provider. Some VPN companies log and sell your data!

- Only use VPNs from trusted, paid providers with clear 'no-logs' policies
- Free VPNs often make money by selling your browsing data
- A VPN is not a magic privacy shield — it's one layer of protection



Think About It!

You use a free VPN to visit a website privately. Is your data really private? Who might still be able to see what you do online?



PASSWORDS, MFA & PASSKEYS

Your first line of defence — make it strong!



Rules for a Strong Password

- At least 12 characters long (longer = stronger!)
- Mix of uppercase, lowercase, numbers AND symbols (#@\$%^&*)
- No real words, names, birthdays, or pet names
- Never reuse the same password on multiple sites
- Use a password manager to remember them all securely

✗ Weak Passwords

password123
john1990
iloveyou



✓ Strong Passwords

P!zzA-T0wer\$Blu3
Tr3e#Jump*Sky9@
M0nkey!L3af_2024^



MFA — Multi-Factor Authentication

MFA adds an extra step after your password. Even if someone steals your password, they still can't get in!



Something you HAVE

A code sent to your phone (SMS), or generated by an app (Google Authenticator, Authy)



Something you ARE

Your fingerprint, face recognition, or iris scan



Passkeys — The Future of Passwords

Passkeys replace passwords entirely! They use your device's biometrics (face or fingerprint) to log you in securely. No password to steal, no password to forget!

- Already supported by Google, Apple, Microsoft and many websites
- Automatically phishing-resistant — no passwords to trick you with!
- Stored on your device, not on a company's server



Think About It!

Your email uses just a password. You activate MFA. How much harder is it now for a hacker to access your account?



AI & DEEPFAKES

When you can't trust your eyes and ears anymore

What are Deepfakes?

Deepfakes are AI-generated photos, videos, or audio clips that look completely real but are 100% fake. AI can swap faces, clone voices, and create fake videos of real people saying or doing things they never did.

The Data Connection — This is Important! Here is something many people don't realize:



Where does training data come from?

Many fun AI apps (haircut changers, face swappers, aging filters, ear color changes...) collect your photos to train their AI models. Some sell this data to third parties.



Tightly linked to Data Privacy

Once your face data is uploaded to these apps, YOU lose control. That data could be used to create deepfakes of you — without your consent.



Sold by companies

Some platforms providing 'fun AI tricks' are actually harvesting your biometric data (face, voice) and selling it. Read the terms and conditions before uploading your face!

How Deepfakes Can Be Used to Harm

- Fake videos to embarrass, bully, or blackmail someone
- Cloned voices to trick family members ('grandparent scam')
- Fake news and misinformation spreading on social media
- Identity theft using your face

How to Spot a Deepfake

- Look for unnatural blinking or facial movements
- Check if hair, teeth, or ears look blurry or distorted
- Background details may flicker or look wrong
- Verify shocking content on trusted news sources before sharing
- Use deepfake detection tools online



Think About It!

A friend sends you a video of a celebrity saying something shocking. Before sharing it, what 3 steps would you take to check if it is real?



APP vs BROWSER

Which one accesses more of your phone?

The Key Difference

When you access a service through an App vs a Browser, the level of access they have to your phone is very different. Understanding this can protect your privacy!



MOBILE APP

Often requests many permissions upfront:

- Camera, microphone, contacts
- Location (even in background)
- Photos & files on your device
- Notifications, calendars, health data



BROWSER

More limited access by default:

- Must ask permission EACH time
- No background location access
- Cannot access your contacts/files
- Sandboxed — separated from your OS



The Golden Rule

With an APP: it can keep access to your camera, location, microphone... even when you're not using it — unless you manually revoke permissions in settings.

With a BROWSER: it typically asks for permission each time, and the access stops when you close the tab.



Smart App Permission Habits

- Review all app permissions in your phone's Settings regularly
- Ask yourself: does this app really NEED this access?
- A flashlight app should NOT need your contacts — that's a red flag!
- Prefer browser access when you only occasionally use a service
- On iPhone: use 'Ask Every Time' for camera & location
- On Android: use 'Only while using the app' for location



Prefer APP when...

You use it daily (navigation, music)
Offline access is needed
Performance matters (games, video)



Prefer BROWSER when...

It is a service you use rarely
You don't want to share location/contacts
You are testing or browsing quickly



Think About It!

You download a free recipe app. It asks for access to your contacts, microphone, and location. Does it need all of that to show you recipes? What would you do?